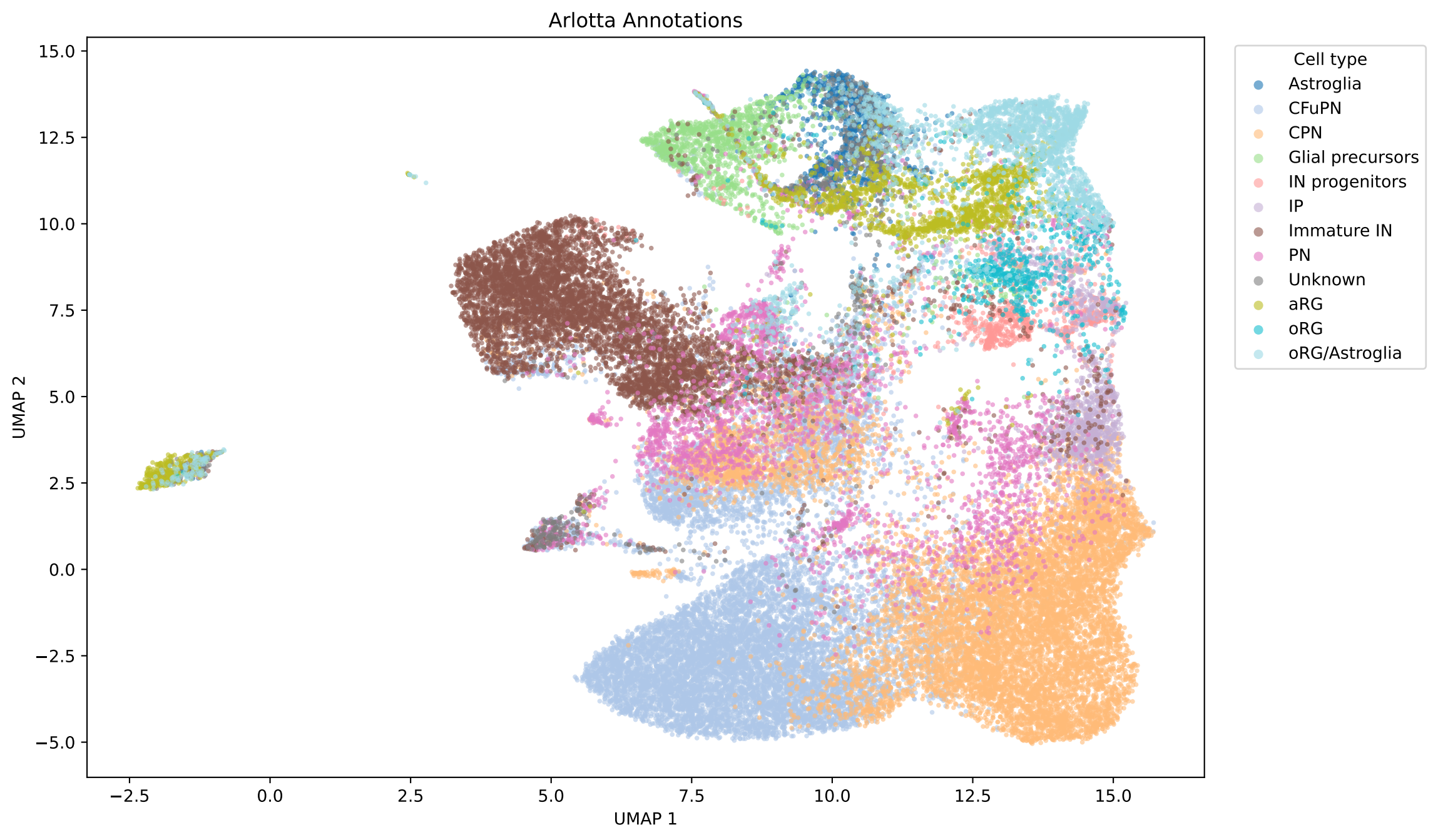
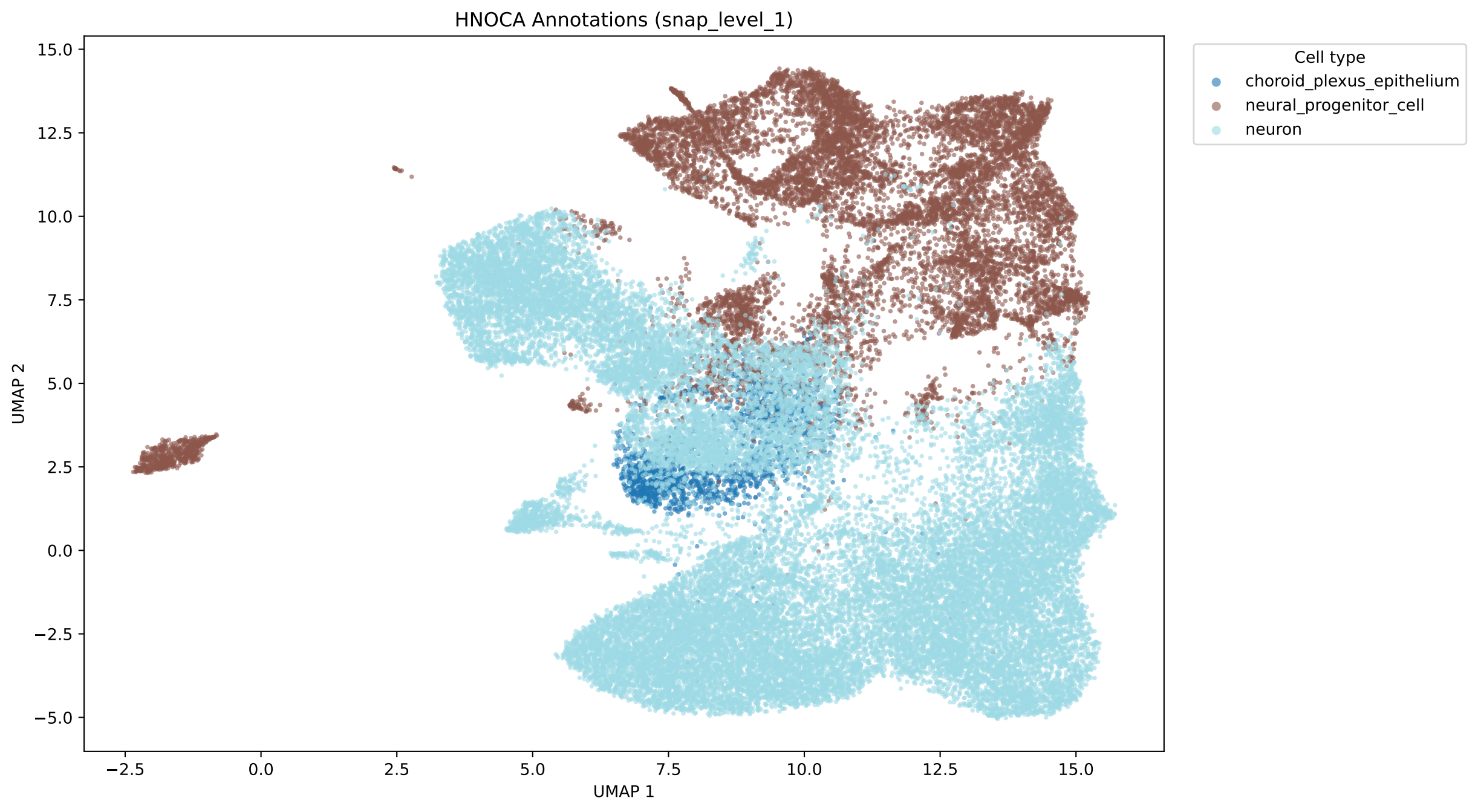
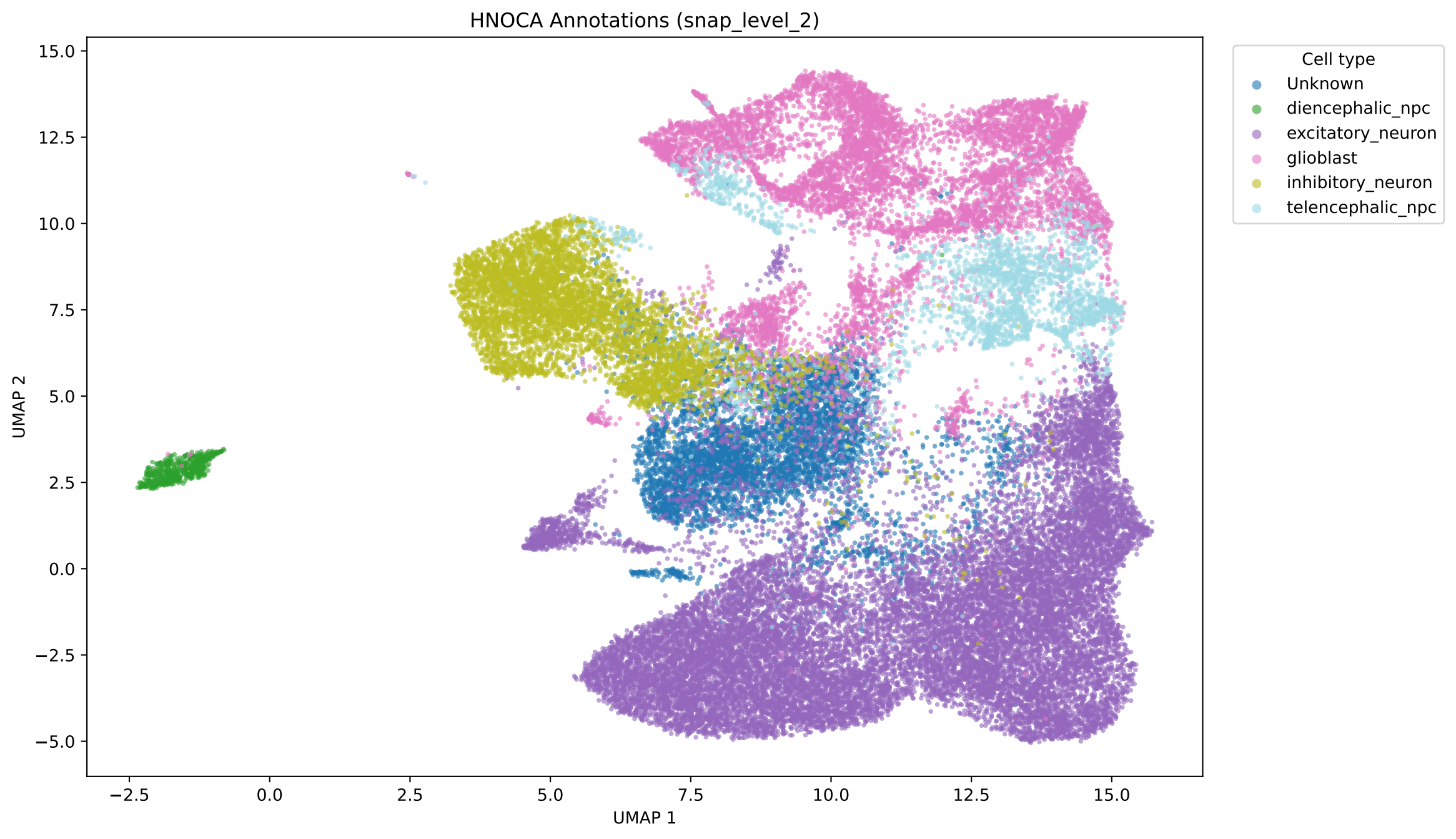
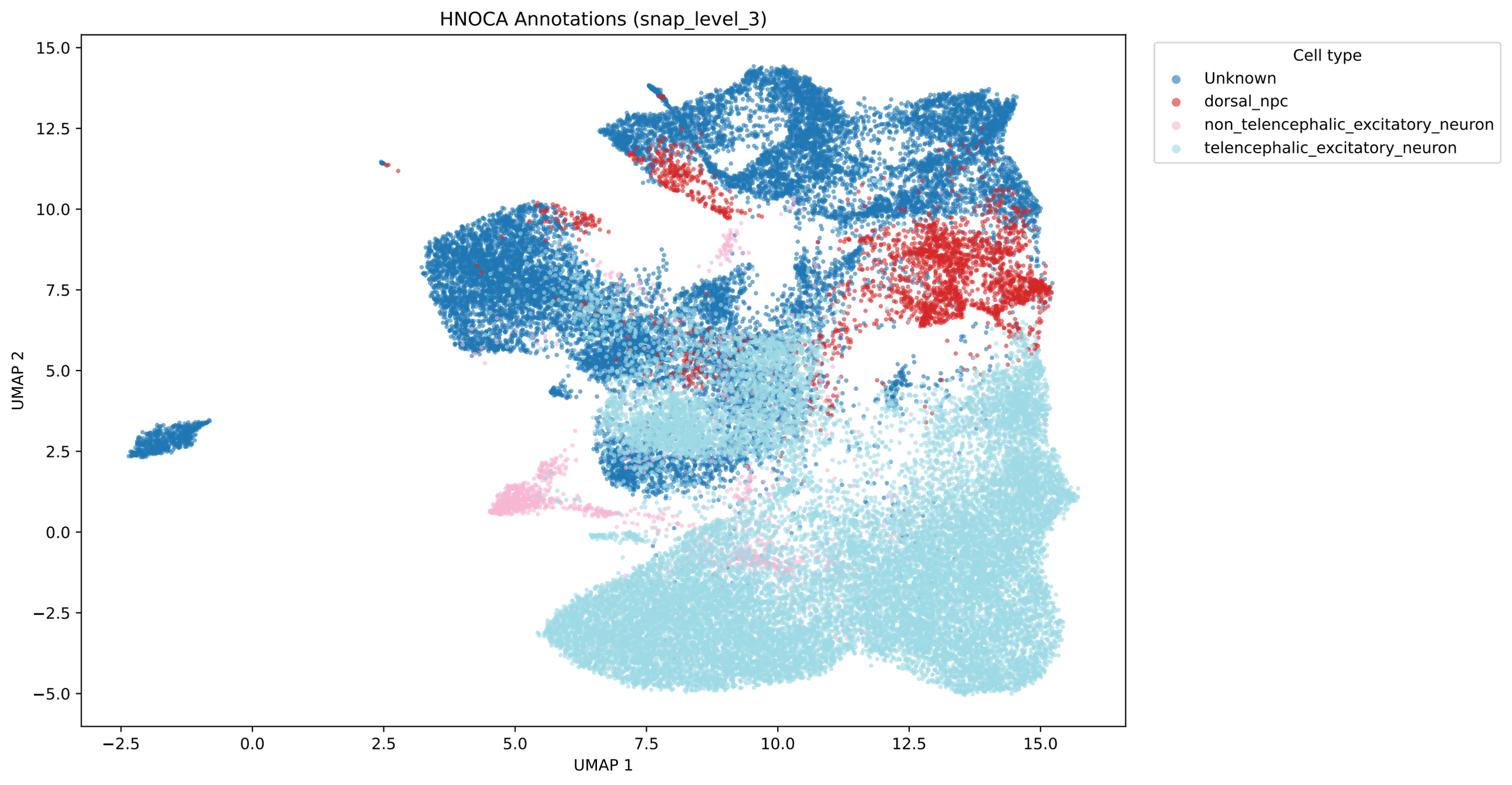
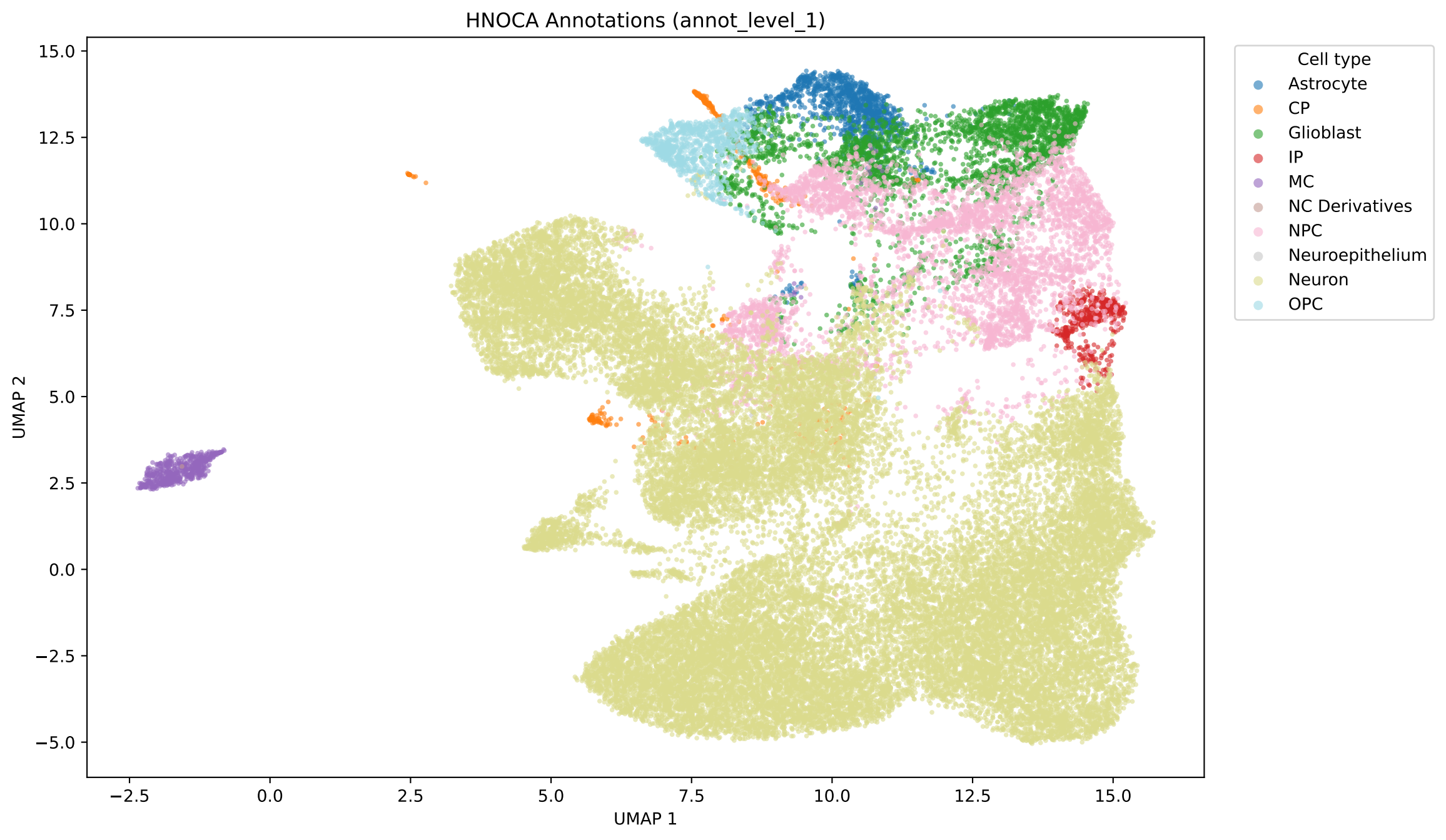
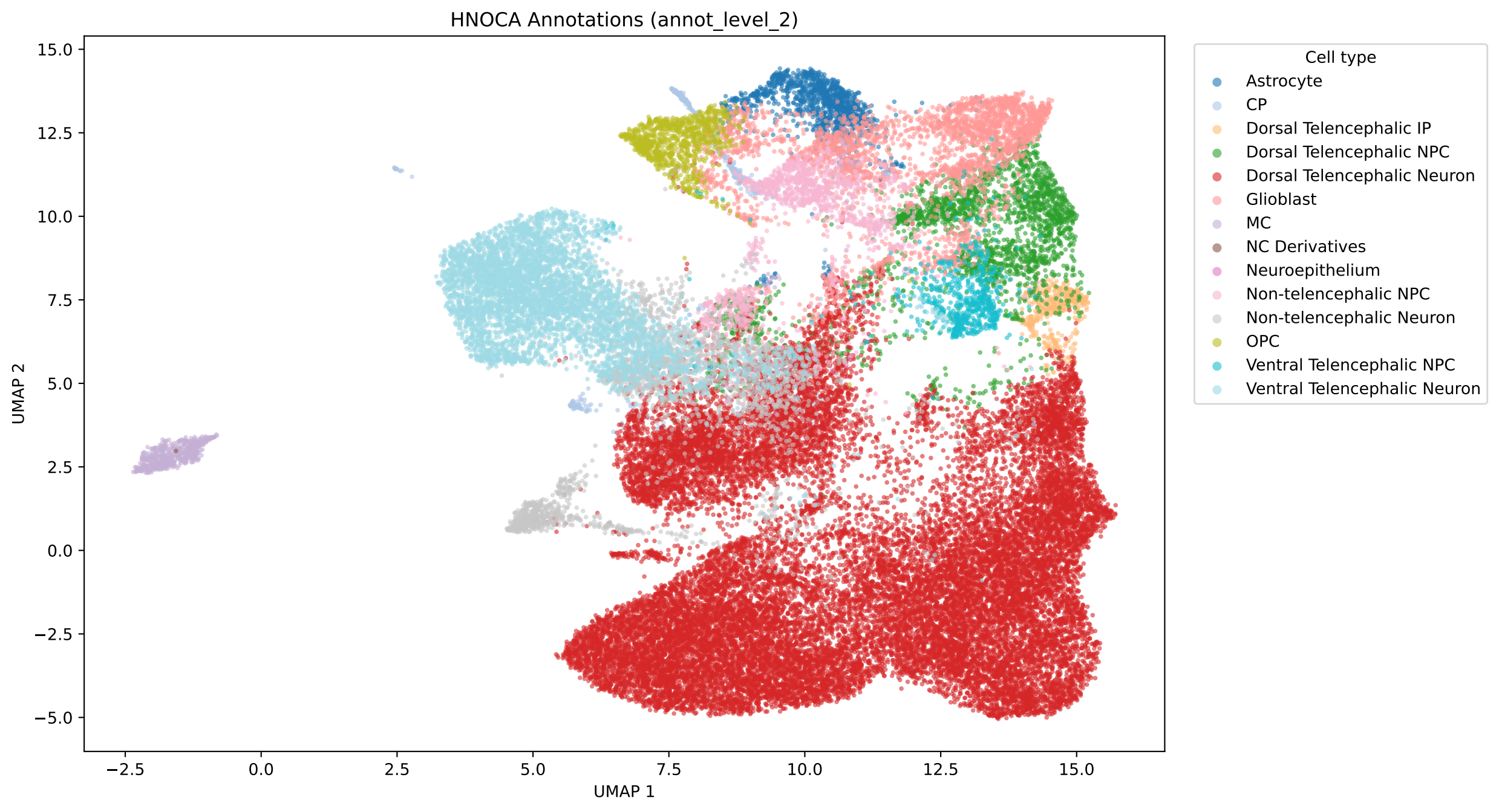

Arlotta Annotations
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Cell type
Astroglia
CFuPN
CPN
Glial precursors
IN progenitors
IP
Immature IN
PN
Unknown
aRG
oRG
oRG/Astroglia

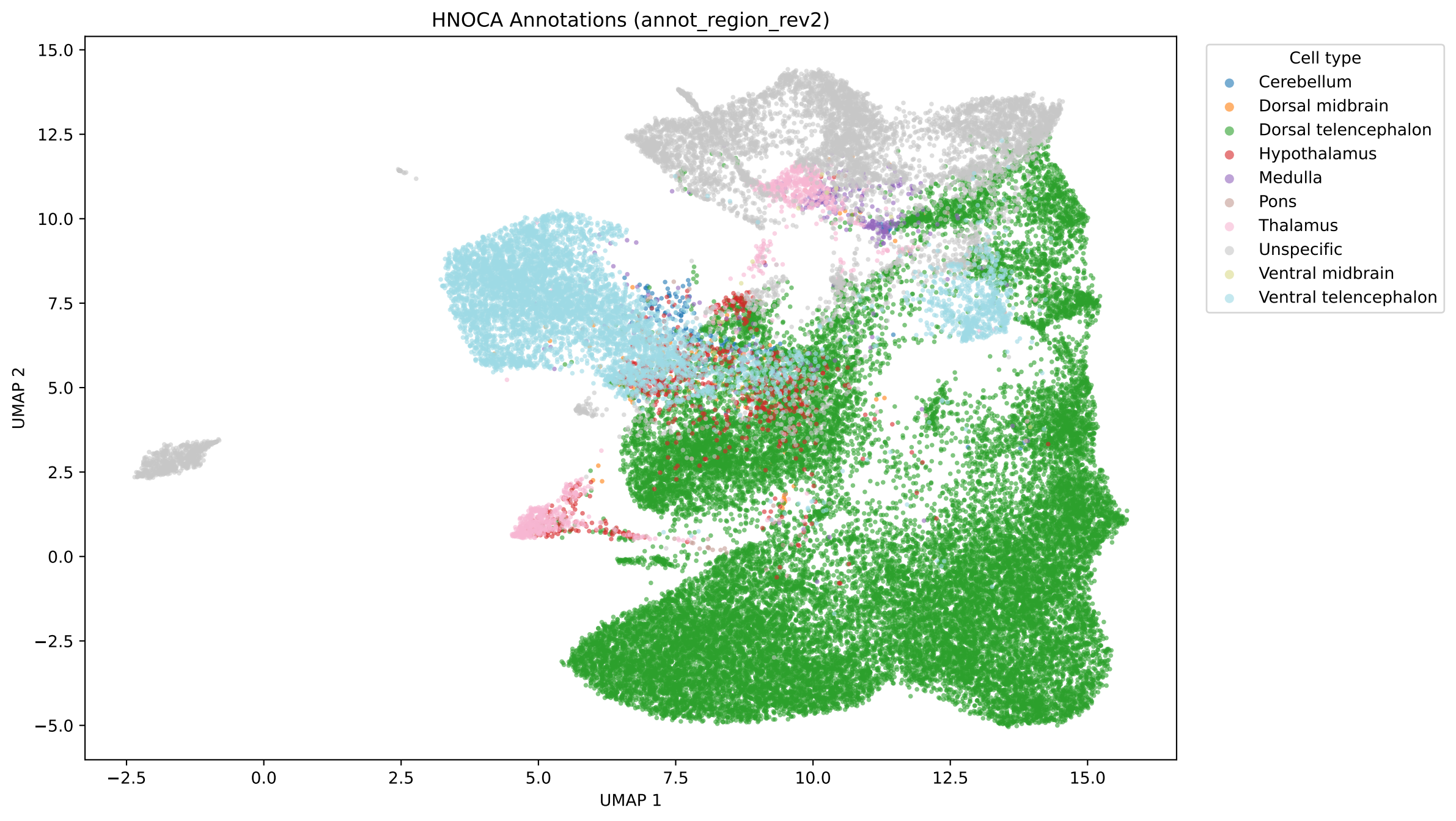

HNOCA Annotations (snap_level_1)
Cell type
choroid_plexus_epithelium
neural_progenitor_cell
neuron
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0

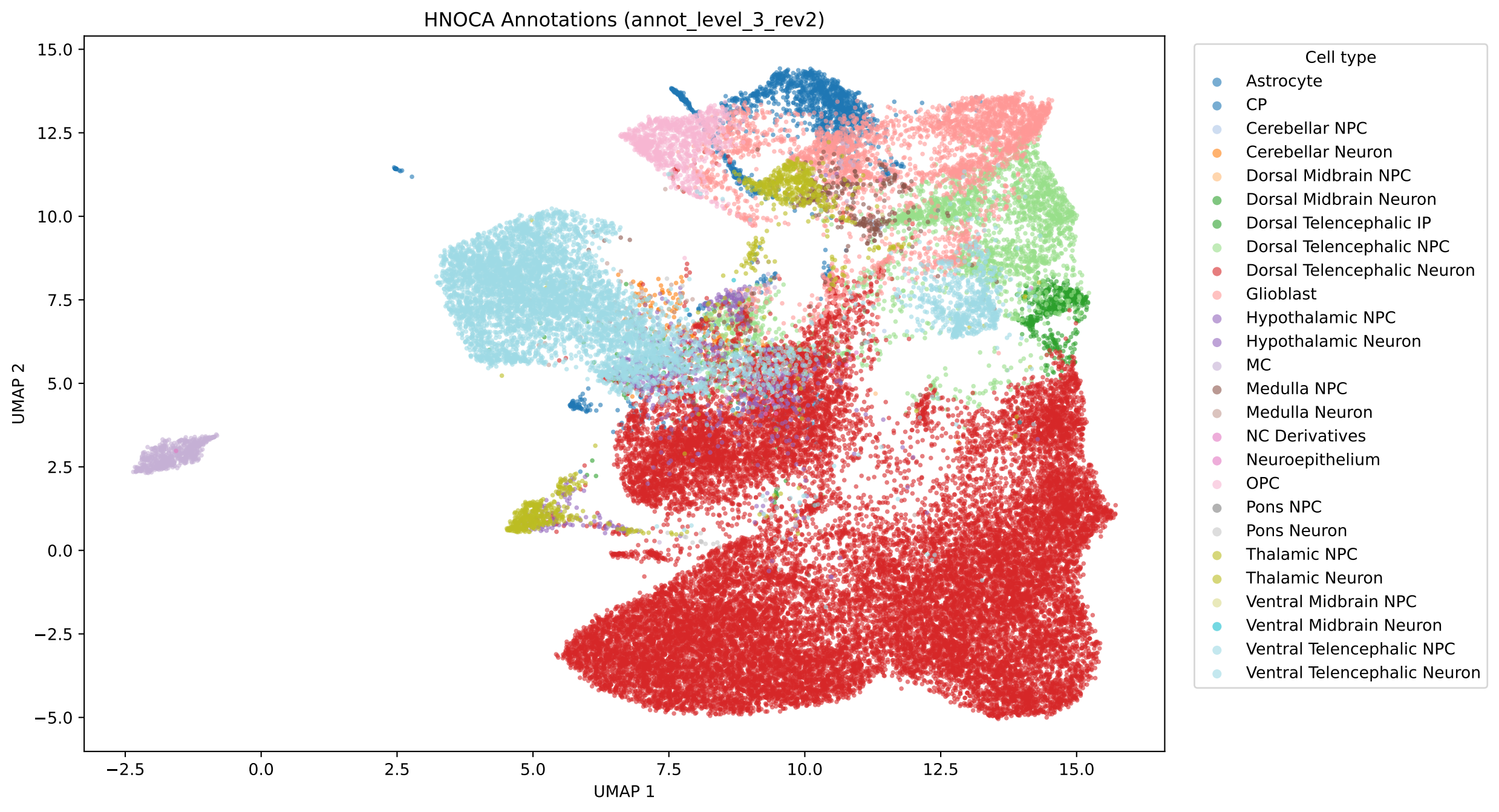

HNOCA Annotations (snap_level_2)
Cell type
Unknown
diencephalic_npc
excitatory_neuron
glioblast
inhibitory_neuron
telencephalic_npc
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0

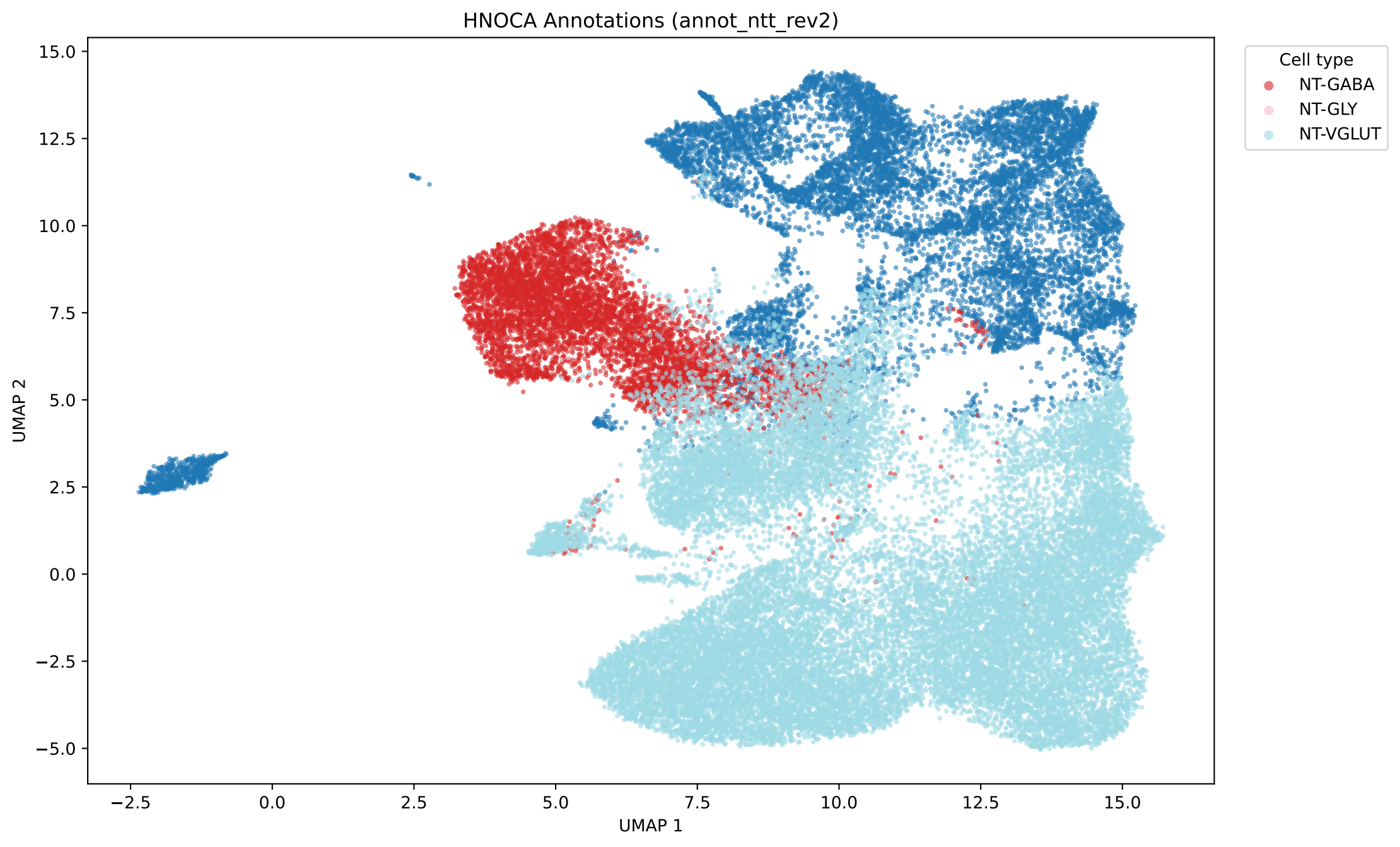

HNOCA Annotations (snap_level_3)
Cell type
Unknown
dorsal_npc
non_telencephalic_excitatory_neuron
telencephalic_excitatory_neuron
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0

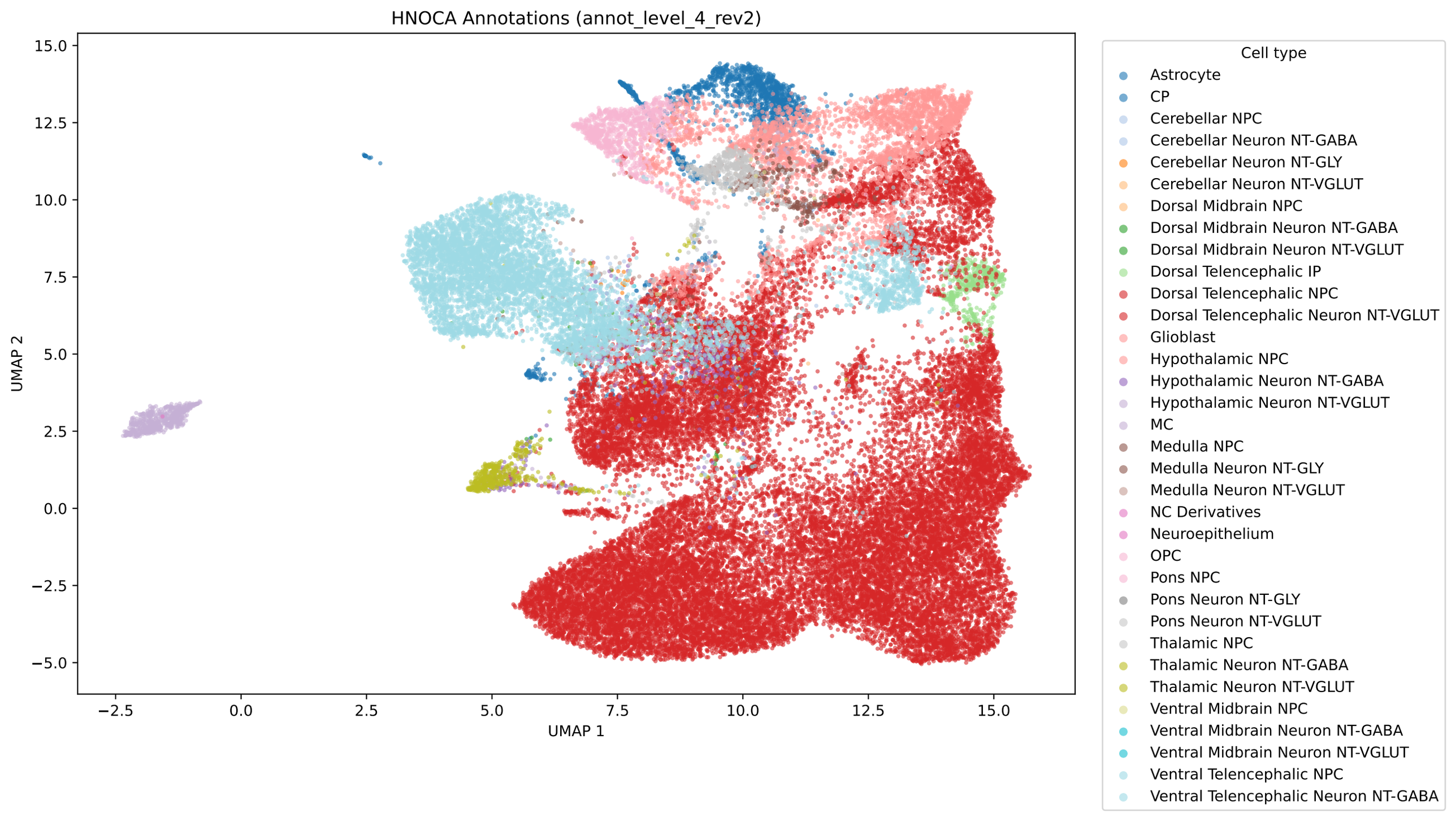

HNOCA Annotations (annot_level_1)
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Cell type
Astrocyte
CP
Glioblast
IP
MC
NC Derivatives
NPC
Neuroepithelium
Neuron
OPC

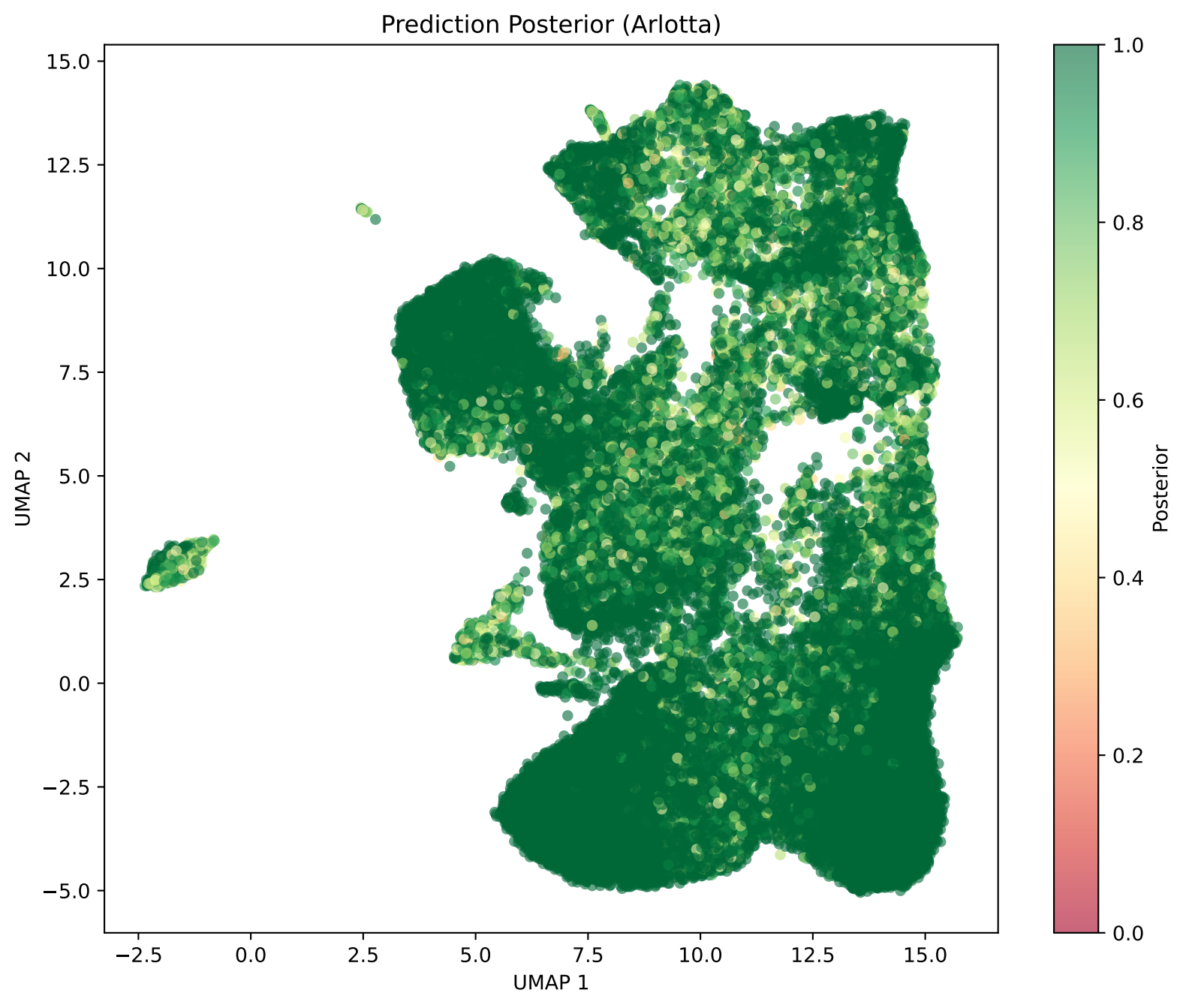

HNOCA Annotations (annot_level_2)
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Cell type
Astrocyte
CP
Dorsal Telencephalic IP
Dorsal Telencephalic NPC
Dorsal Telencephalic Neuron
Glioblast
MC
NC Derivatives
Neuroepithelium
Non-telencephalic NPC
Non-telencephalic Neuron
OPC
Ventral Telencephalic NPC
Ventral Telencephalic Neuron


HNOCA Annotations (annot_region_rev2)
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Cell type
Cerebellum
Dorsal midbrain
Dorsal telencephalon
Hypothalamus
Medulla
Pons
Thalamus
Unspecific
Ventral midbrain
Ventral telencephalon


HNOCA Annotations (annot_level_3_rev2)
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Cell type
Astrocyte
CP
Cerebellar NPC
Cerebellar Neuron
Dorsal Midbrain NPC
Dorsal Midbrain Neuron
Dorsal Telencephalic IP
Dorsal Telencephalic NPC
Dorsal Telencephalic Neuron
Glioblast
Hypothalamic NPC
Hypothalamic Neuron
MC
Medulla NPC
Medulla Neuron
NC Derivatives
Neuroepithelium
OPC
Pons NPC
Pons Neuron
Thalamic NPC
Thalamic Neuron
Ventral Midbrain NPC
Ventral Midbrain Neuron
Ventral Telencephalic NPC
Ventral Telencephalic Neuron


HNOCA Annotations (annot_ntt_rev2)
Cell type
NT-GABA
NT-GLY
NT-VGLUT
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0


HNOCA Annotations (annot_level_4_rev2)
UMAP 1
UMAP 2
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Cell type
Astrocyte
CP
Cerebellar NPC
Cerebellar Neuron NT-GABA
Cerebellar Neuron NT-GLY
Cerebellar Neuron NT-VGLUT
Dorsal Midbrain NPC
Dorsal Midbrain Neuron NT-GABA
Dorsal Midbrain Neuron NT-VGLUT
Dorsal Telencephalic IP
Dorsal Telencephalic NPC
Dorsal Telencephalic Neuron NT-VGLUT
Glioblast
Hypothalamic NPC
Hypothalamic Neuron NT-GABA
Hypothalamic Neuron NT-VGLUT
MC
Medulla NPC
Medulla Neuron NT-GLY
Medulla Neuron NT-VGLUT
NC Derivatives
Neuroepithelium
OPC
Pons NPC
Pons Neuron NT-GLY
Pons Neuron NT-VGLUT
Thalamic NPC
Thalamic Neuron NT-GABA
Thalamic Neuron NT-VGLUT
Ventral Midbrain NPC
Ventral Midbrain Neuron NT-GABA
Ventral Midbrain Neuron NT-VGLUT
Ventral Telencephalic NPC
Ventral Telencephalic Neuron NT-GABA


Prediction Posterior (Arlotta)
UMAP 2
UMAP 1
Posterior
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
1.0
0.8
0.6
0.4
0.2
0.0